MILITARY SPECIFICATION

TELEPHONE EQUIPMENT, DIAL (SHIPBOARD USE)

This amendment forms a part of MIL-T-1943C(SHIPS), dated 31 May 1968, and is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2 Telephone station equipment: After "Relays for extension signals" add "Telephone sets for use on restricted lines (see 3.6.10.1) shall be designated by '(R)' after the type designation letter."

PAGES 1 and 2

2.1: Add the following Military standards:

"MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
"MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
"MIL-STD-1310 - Shipboard Bonding and Grounding Methods for Electromagnetic Compatibility."

PAGE 2

* 3.1: Delete and substitute:

"3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2)."

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PAGES 2, 3, and 4

3.2, listing: Make the following changes:

(a) After "Enclosures - general" delete the referenced paragraphs.
(b) After "Enclosure - degree of" add "see 3.6.1.1.1, 3.6.1.2, 3.6.1.2.1, 3.6.1.4.1, 3.6.1.5, 3.6.1.5.1, 3.6.1.5.3, 3.6.2.1.1, 3.6.9, 3.7.7.4, 3.7.8.1, 3.7.8.2, 3.8.2, 3.8.3.2, 3.8.4.2, 3.8.4.4, and 3.9.1.2".
(c) After "Shock, vibration (vital) and inclination" delete "(see 3.7.7.1)" and substitute "(see 3.4 and 3.7.7.1)".
(d) After "Shielding and radio frequency noise reduction" add "(see 3.19, 3.19.1 and 4.5.12)".
(e) After "Ground potential and grounding" add "(see 3.20)".
(f) After "designation and marking" add "(see 3.18)".

PAGE 5

3.6.1.3: Delete and substitute:

"3.6.1.3 Type D. The type D telephone set is a special intercommunicating type of telephone set. The details of design, dielectric strength, insulation resistance, and other similar requirements shall be specified (see 6.1)."

PAGE 6

3.6.1.5.2: Delete and substitute:

"3.6.1.5.2 Desk enclosure. The desk enclosure for a type G telephone set shall be in accordance with 3.6.1.1.1 except with regard to shape, mounting, and color. The enclosure shall not exceed 6 inches wide by 12 inches high, but shall mount the panel on a slant by means of a 5-inch end and a 2-1/4-inch end. For mounting a desk, three threaded inserts shall be provided in the base and in the larger end. Attachment shall be made by bolts screwed into the inserts from the under side of the mounting surface. The enclosure shall be suitable for holding the set when the enclosure is base or end mounted. The enclosure shall be painted gray in accordance with MIL-I-983. The enclosure shall be fitted with a 6-foot desk set cord and a nonwatertight connection block and cover in accordance with 3.6.1.1.2."

PAGES 6 and 7

3.6.2.1, first line: Between "and" and "receiver", insert "a".

PAGE 8

* 3.6.7: Delete and substitute:

"3.6.7 Handset holding and line switch actuating mechanisms. The design of all telephone handset holding and line switch actuating mechanisms shall be subject to specific approval by the Naval Sea Systems Command."

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3.6.10.1: Delete and substitute:

"3.6.10.1 Telephone sets for restricted lines. Telephone sets to be installed on restricted lines shall have distinctive identification plates (see 3.18.2) and shall be equipped with special handsets having noise canceling microphones and shielded cords (see 3.6.2.1 and 3.6.6) as specified (see 6.1)."

3.7, line 2: After "telephone lines" insert "(see 6.4.8)."

PAGE 9

3.7.3: Delete and substitute:

"3.7.3 Switchboard circuitry."

3.7.3.2, first sentence: Delete and substitute "Switchboard circuitry shall provide for the maximum feasible number of connection paths or links between calling and called lines, and shall be arranged within each line finder group so that a blocked call does not prevent other calls from being established."

Add new paragraph 3.7.3.4:

"3.7.3.4 In switchboards of 200-line and larger capacities, provision shall be made for expansion, in 100-line increments, to larger capacities. It shall be possible to effect an expansion by the addition of 100-line switchboard sections, and modification of interconnections, including suitable selector grading changes, at readily available terminal boards."

3.7.4: Delete and substitute:

"3.7.4 Dialing. In switchboards of less than 100-line capacity, two-digit dialing shall be used, except that in the 40-line switchboard, additive dialing not requiring over five digits may be used. In switchboards of 100-line capacity and larger, three-digit dialing shall be used, and all ten digits shall be available for assignment as the first digit of directory numbers, distributed approximately equally between the 100-line groups available. In order to permit the installing activity to proceed with the preparation of the telephone directories and isometric wiring diagrams of the telephone system in advance of the approval of equipment drawings, information of the numbering scheme, including the method of assigning party-line numbers, shall be furnished to the procuring and installing activities concerned within 30 days after the award of the contract or order."

PAGE 10

3.7.5.3, line 1: Delete "MIL-S-15303" and substitute "MIL-A-15303".

PAGE 12

3.7.7.19: Delete and substitute:

"3.7.7.19 Test circuit. The switchboard or each 100-line group shall contain a suitable test circuit to facilitate routine testing of the switchboard equipment. Connector terminal 29 shall be assigned to this circuit."

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3.7.7.23: Delete and substitute:

"3.7.7.23 Line tests. Provision shall be made to permit operating personnel on board ship to make tests of individual telephone line readily and easily. In step-by-step type switchboards, this requirement is met by the semi-automatic line tester (see 3.10.2(c))."

PAGE 13

3.7.9.3: Delete and substitute:

"3.7.9.3 Single and two-party service. All switchboards, except the 40-line switchboard, shall be designed for single and two-party line service. The switchboard circuitry shall provide for ringing over the line pair for single party service or for individual ringing over one side or the other side of the line and a common return wire, in accordance with dialed digits, for two-party service. In the 25, 50, 100, 200, 300, 400, and 500-line switchboards, all lines shall be for either single or two-party service. In the 150-line switchboard, the first 100 lines shall be single party lines and the other 50 lines shall be two-party lines. In the 600-line switchboard, single party lines shall be provided in the two 100-line groups and single and two-party lines shall be provided in the other four 100-line groups. In the 700-line switchboard, single party lines shall be provided in the four 100-line groups and single or two-party lines shall be provided in the other three 100-line groups."

3.7.9.4: In the first line, delete "0" and substitute "11"; in the fifth line, between "dialing" and "211", insert "11 or".

PAGE 14

Add new paragraph 3.7.9.6:

"3.7.9.6 Isolated 100-line switchboard section. When specified (see 6.1), for the purpose of segregating a group of stations, such as aircraft maintenance stations of an aircraft carrier, a 100-line section of a 300-line or larger switchboard shall be wired to provide maximum and partially exclusive use of its connectors to calls initiated in that 100-line section. The isolated section shall be a mid-section of the 100-line sections of the switchboard. In the isolated section, ten of the fifteen connectors shall be available only to selectors of the isolated section. The remaining five connectors shall be available to selectors of all of the 100-line sections."

PAGE 15

3.8.2: In the third line, between "aluminum" and "cabinet", insert "dripproof".

PAGE 17

3.9.1.5: At the end of the paragraph add "except as specified in 3.7.9.5."

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PAGES 18 and 19

Table I: Under "Switchboard" delete the listings for "Relays each regular type" and "Relays each special type" and substitute the following:

	Size of switchboard (lines)						
	25	40 and 50	100	150	200	300	400 and larger
"Relays each regular type (except those furnished as an integral part of a regular automatic switch unit)	2	4	8	9	10	12	14"
"Relays each special type (except those furnished as an integral part of a special automatic switch unit	1	1	1	1	1	1	1"

PAGE 21

Add new paragraphs 3.18 through 3.18.2, 3.19, 3.19.1 and 3.20:

"3.18 Designation and marking. All units of dial telephone equipment shall be provided with identification plates in accordance with MIL-I-983 and shall be type A or B of MIL-P-15024.

"3.18.1 The information on identification plates shall include the following:

'Dial telephone equipment.
Unit identification such as telephone type A or switchboard 100-line.
Stock number when contract is for stock.
Department of the Navy - Naval Sea Systems Command.
Manufacturer's name and part or drawing number.
Contract or purchase order number on switchboard, power, and accessory equipment.
A number to permit identifying the date of manufacture by month and year on telephone station equipment.

"3.18.2 Identification plates for telephone sets to be installed on restricted lines shall have 'RESTRICTED' as the first line, marked as large as practicable. Also, '(R)' shall be added after the type designation.

"3.19 Shielding and radio frequency noise reduction. The design requirements for the dial telephone equipment shall be in accordance with MIL-I-983, except that the limits shall be as specified in MIL-STD-461. Switchboards are designated as class ID; motor-generator type power equipment is designated class IIB.

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"3.19.1 Bonding. The requirements of MIL-I-983 shall apply, except that the bonding methods shall be in accordance with MIL-STD-1310.

"3.20 Ground potential and grounding. The requirements of MIL-I-983 shall apply except that bonding shall be in accordance with MIL-STD-1310."

* 4.2.1: Delete and substitute:

"4.2.1 First article inspection. First article inspection shall consist of the examinations and tests specified in 4.2.1.1."

* 4.2.1.1, Table II: Delete "Qualification" and substitute "First Article".

Table II: Delete the listings for "Shielding and radio frequency noise reduction" and "Enclosure" and substitute the following:

Examination or test	Requirement paragraph	Examination or test paragraph
"Shielding and radio frequency noise reduction	3.2, 3.19	4.5.12"
"Enclosure	3.6.1.1.1, 3.6.1.2, 3.6.1.2.1, 3.6.1.4.1, 3.6.1.5, 3.6.1.5.1, 3.6.1.5.3, 3.6.2.1.1, 3.6.9, 3.7.7.4, 3.7.8.1, 3.7.8.2, 3.8.2, 3.8.3.2, 3.8.4.2, 3.8.4.4 and 3.9.1.2	4.5.5"

* 4.2.1.1, Table II: Delete "footnote 1/" in its entirety.

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* Table II: Delete "Qualification tests" and substitute "First article tests".

Table III: Delete the listings for "Shielding and radio frequency noise reduction", "Enclosure" and "Vibration" and substitute the following:

Inspection	Requirement paragraph	Examination or test paragraph
"Shielding and radio frequency noise reduction	3.2, 3.19	4.5.12"
"Enclosure	3.6.1.1.1, 3.6.1.2, 3.6.1.2.1, 3.6.1.4.1, 3.6.1.5, 3.6.1.5.1, 3.6.1.5.3, 3.6.2.1.1, 3.6.9, 3.7.7.4, 3.7.8.1, 3.7.8.2, 3.8.2, 3.8.3.2, 3.8.4.2, 3.8.4.4 and 3.9.1.2	4.5.5"
"Vibration	3.2, 3.7.7.1	4.5.6.1"

PAGE 23

* 4.3.4, first sentence: Delete and substitute "Group C tests will be performed when required by the Naval Sea Systems Command."

PAGE 28

* Add new paragraph 4.5.12:

"4.5.12 Radio frequency noise interference. Radio frequency noise interference test shall be performed as specified in MIL-STD-462 (see 3.19)."

PAGE 30

* 6.1: Add items (o) and (p):

"(o) Whether an isolated 100-line switchboard section is required (see 3.7.9.6).
"(p) First article inspection, if required (see 3.1)."

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* 6.2: Delete and substitute:

"6.2 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract."

* 6.3: Delete in its entirety.

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6.4.7: Delete and substitute:

"6.4.7 Standard handset. A standard handset is one designed for general use as a part of the telephone sets of a telephone system, except as specified in 3.6.10.1."

* 6.5: Delete in its entirety.

LAST PAGE

DD Form 1426, Standardization Document Improvement Proposal: Delete address and substitute:

"COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101"

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